Name: ______________________

VECTORS, (MULTIPLYING VECTORS BY A SCALAR)

Date: ________________

LO: I can multiply a vector by a scalar and understand the geometric effect.

Scalar multiplication

A **scalar** is just a number (no direction). Multiplying a vector by a scalar **scales** its magnitude. The direction stays the same (or reverses if negative).

Rule: k(a, b) = (ka, kb). Multiply each component by the scalar.

Quick examples

●	3(2, 4) = (6, 12)	multiply both by 3
●	-2(1, 3) = (-2, -6)	reverses direction
●	½(6, 4) = (3, 2)	halves the vector
●	3(2, 4) = (6, 4)	must multiply both
●	-1(3, 2) = (3, 2)	should be (-3, -2)

Key idea

k(a, b) = (ka, kb). Positive k keeps direction; negative k reverses it.

k (a, b) = (ka, kb)

Scale both components by the same factor

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – positive scalar

Find 3(2, 5).

Multiply x: 3 × 2 = 6

Multiply y: 3 × 5 = 15

= (6, 15)

(6, 15)

Multiply each component by the scalar.

Example 2 – negative scalar

Find -2(4, -1).

-2 × 4 = -8

-2 × (-1) = 2

= (-8, 2)

(-8, 2)

Negative scalar reverses direction.

Example 3 – fractional scalar

Find ½(8, -6).

½ × 8 = 4

½ × (-6) = -3

= (4, -3)

(4, -3)

Fractional scalar reduces the magnitude.

Example 4 – parallel vectors

Show that (4, 6) is parallel to (2, 3).

(4, 6) = 2(2, 3)

One is a scalar multiple of the other

So they are parallel

Parallel: (4,6) = 2 × (2,3)

Parallel vectors are scalar multiples of each other.

YOUR TURN – PRACTICE (deliberate practice)

1. Positive scalars

Calculate each scalar multiplication.

a) 2(3, 4)

b) 4(1, -2)

c) 5(2, 0)

d) 3(-1, 5)

2. Negative scalars

Calculate each.

a) -1(3, 7)

b) -3(2, -4)

c) -2(-5, 1)

d) -4(1, 2)

3. Fractional scalars

Calculate each.

a) $\frac{1}{2}$(4, 10)

b) $\frac{1}{3}$(9, -6)

c) $\frac{2}{3}$(6, 9)

d) $\frac{1}{4}$(8, -12)

4. Identifying parallel vectors

Are these vectors parallel? If yes, state the scalar.

a) (2,3) and (6,9)

b) (4,1) and (8,3)

c) (1,-2) and (-3,6)

d) (5,10) and (1,2)

5. Combined operations

Calculate.

a) 2(3,1) + (4,2)

b) 3(1,2) - 2(4,-1)

c) $\frac{1}{2}$(4,6) + $\frac{1}{3}$(9,-3)

d) 2a - b where a=(3,-1), b=(2,4)

e) a + 3b where a=(1,5), b=(2,-1)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) k(a, b) = (ka, kb) ☐

Correct answer: __________

Reason: __________________

b) 3(2, 4) = (6, 4) ☐

Correct answer: __________

Reason: __________________

c) Parallel vectors are scalar multiples ☐

Correct answer: __________

Reason: __________________

d) A negative scalar keeps the same direction ☐

Correct answer: __________

Reason: __________________

e) $\frac{1}{2}$a has half the magnitude of a ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Points A, B, C have position vectors **a = (1, 2), b = (4, 5), c = (7, 8)**. Show that A, B, C are collinear by proving AB is parallel to AC.

Expression: ______________________________

Simplified: ______________________________

b) Vector a = **(3, 4)**. Find the magnitude |a|. Find a unit vector in the direction of a (divide by magnitude).

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!